CHAPTER 1 · LESSON 3 OF 5

Sources and Formats of Data

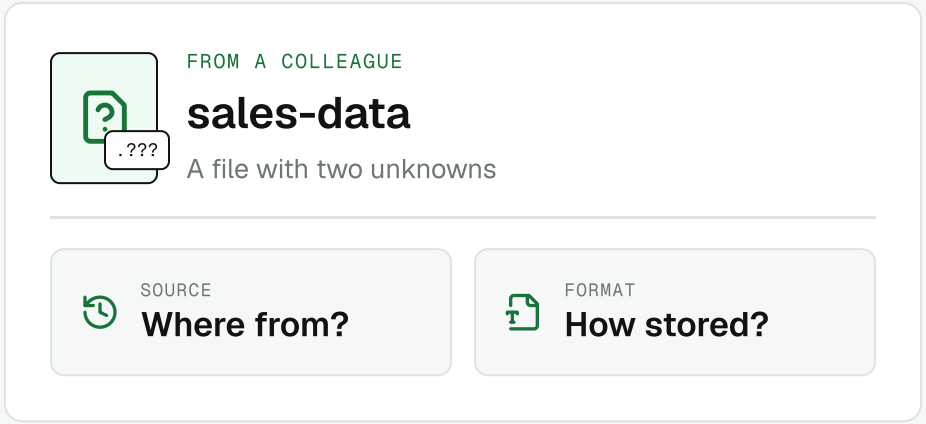
BY THE END

You can identify where a dataset came from, what format it is in, and whether it is primary or secondary.

Overview

You receive a file from a colleague and need to decide whether you can trust it and how to open it.

That decision depends on where the data came from and how it was stored.



Common sources of data

A data source is where data originally comes from. Data can be collected in many ways:

Manual entry

someone types data into a form or spreadsheet

Business systems

sales, inventory, or HR software

Websites and apps

clicks, sign-ups, purchases

Sensors and devices

weather stations, fitness trackers, machines

Surveys and interviews

customer or employee feedback

Public datasets

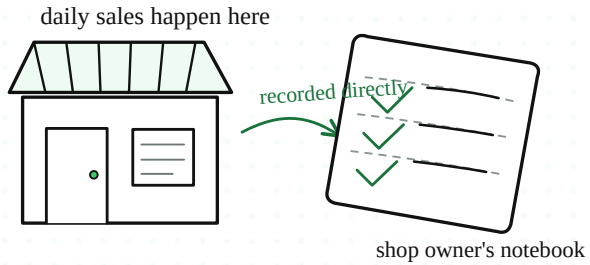
government or research data shared openly

The source affects data quality. Data from a reliable business system may be more consistent than data typed manually into a spreadsheet.

Primary vs secondary data

Primary data

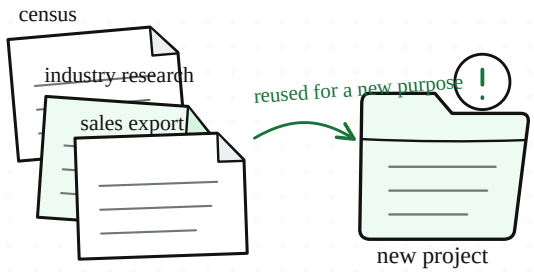
Primary data is collected firsthand for a specific purpose: a shop owner recording daily sales in a notebook, or a school conducting its own student survey.



collected firsthand

Secondary data

Secondary data was collected by someone else and is reused for a new purpose: government census reports, industry research, or sales data exported from another department.



collected elsewhere, reused here

Primary data is often more relevant to your exact question. Secondary data can save time, but you must check whether it fits your needs and whether you trust the original source.

Common data formats

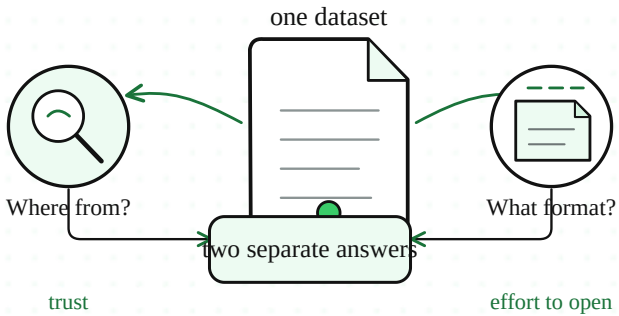
A **data format** is how data is stored or shared.

FORMAT	COMMON USE
CSV	Simple tabular data in text files
Excel (.xlsx)	Spreadsheets with formulas and formatting
JSON	Web and application data exchange
SQL databases	Structured business data stored in tables
PDF / Word	Reports and documents, often harder to analyze directly

As a beginner, you will most often encounter spreadsheets, CSV files, and database tables.

Why source and format matter

Ask two questions about any dataset before you use it.
Where did this come from? tells you whether to trust it.
What format is it in? tells you what it will cost you to open.



The two answers are independent, which is what makes both questions worth asking:

Sales data from a company database usually answers both well - **trusted source, easy to open**

A PDF report can be entirely trustworthy and still **cost you a day to get the numbers out of**

Survey responses can arrive in a clean spreadsheet and still be **full of missing answers**